Counters....

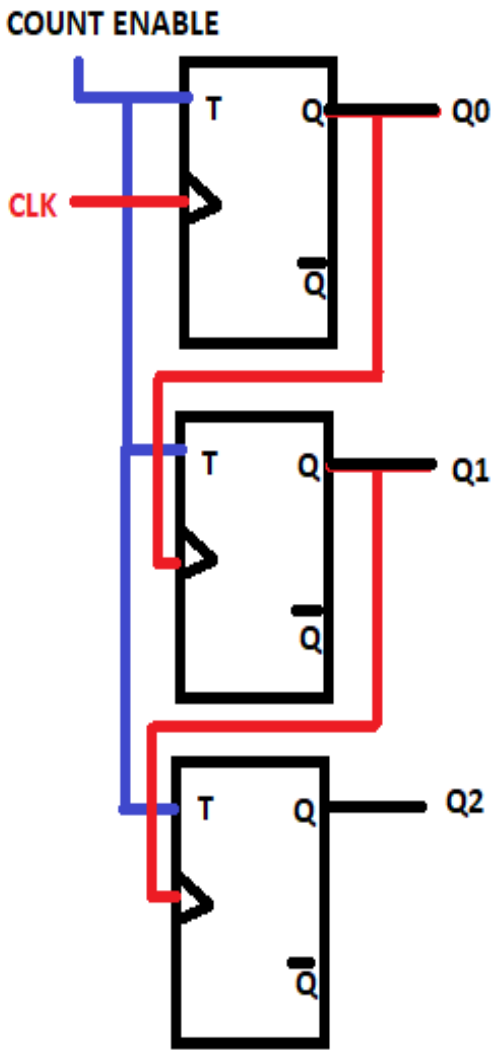
Core Comparison: Synchronous vs. Asynchronous

Feature	Asynchronous (Ripple) Counter	Synchronous Counter
Clock Connection	Cascaded (Output of one drives the next)	Simultaneous (All flip-flops share one clock)
Operating Speed	Slow (Limited by propagation delays)	High (Independent of number of bits)
Circuit Complexity	Simple, fewer logic gates	Complex, requires extra combinational logic
Glitches / Hazards	High risk during transitions	Very low risk (Glitch-free)

Digital counters are sequential logic circuits used to count clock pulses. The modulus (or "Mod") refers to the total number of unique states a counter goes through before resetting to its starting value.

Digital counters are fundamental building blocks in modern electronics. Their applications span from everyday consumer electronics to advanced industrial automation, scientific instrumentation, and microprocessors. Ex: Digital Clocks and Watches, Frequency Dividers & counters. Used in Microprocessors, Computers, and State Machines, used in Industrial Automation and Measuring Instruments, Used in Consumer Electronics and Home Appliances

Mod-8 Asynchronous Counter for Down Counting Using +ve Edge Triggered T Flip-Flops and Using Qi outputs

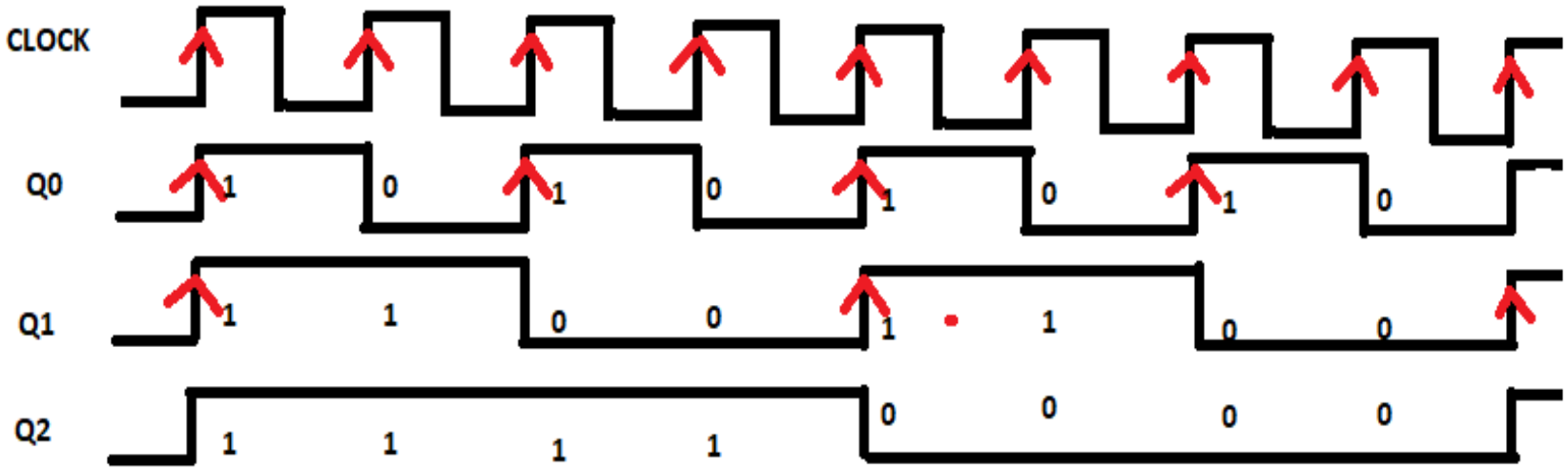


COUNT SEQUENCE

	Q2	Q1	Q0
7	1	1	1
6	1	1	0
5	1	0	1
4	1	0	0
3	0	1	1
2	0	1	0
1	0	0	1
0	0	0	0

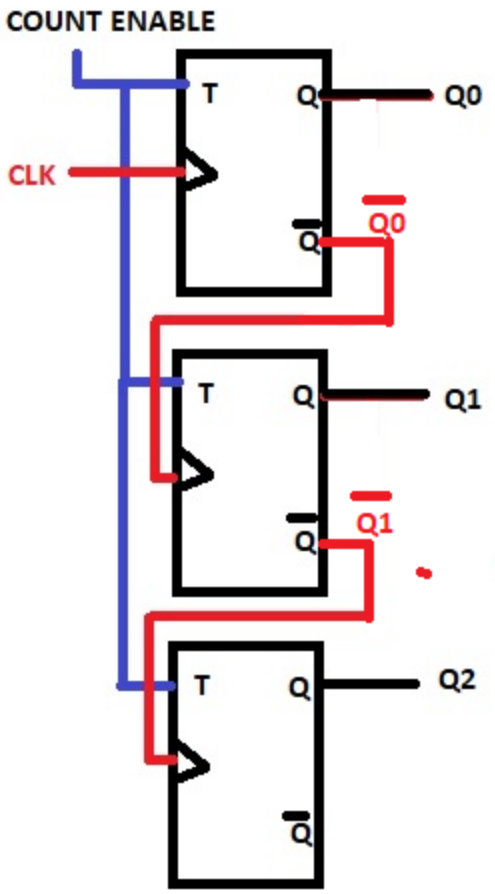
1 1 1

MOD - 8 AYNCHRONOUS COUNTER



1. +VE Edge Triggered T flip-flops are used
2. Input clock is given directly to clock input of Q0 flip-flop
3. Q0 O/P given to CLK input of Q1 flip-flop, Q1 o/p given to CLK input of Q2 flip-flop
4. Count Enable = 1, during Counting (Count Enable = 0, disables the counting)
5. Total States of the counter = 8, Hence Mod-8 Counter
6. As the count sequence is 7->6->5->4->3->2->1->0->7...it is binary down counter

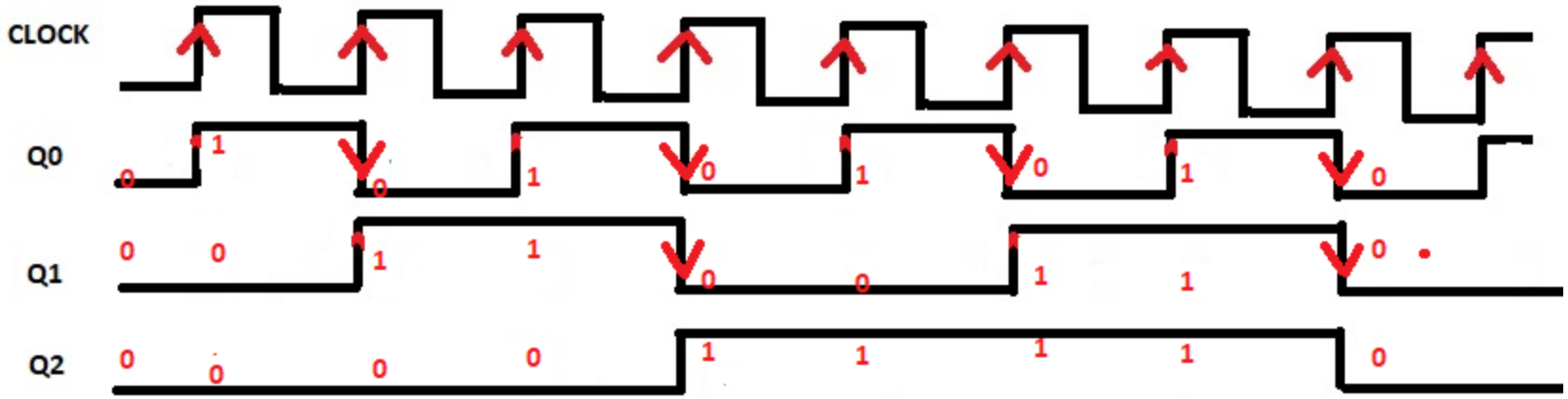
Mod-8 Asynchronous Counter for UP Counting Using +VE Edge Triggered T Flip-Flops and Using Complimentary o/p



COUNT SEQUENCE

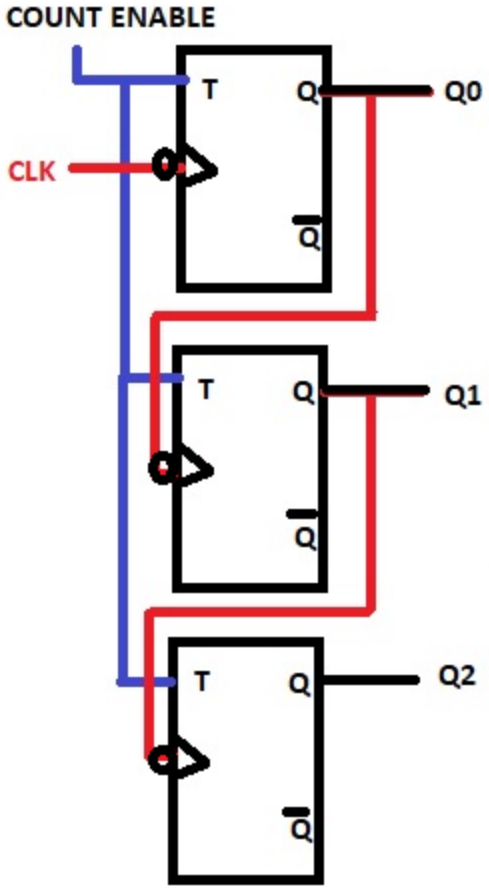
Q2	Q1	Q0
0	0	0
1	0	0
2	0	1
3	0	1
4	1	0
5	1	0
6	1	1
7	1	1
0	0	0

MOD - 8 AYNCHRONOUS COUNTER



1. +VE Edge Triggered T flip-flops are used
2. Input clock is given directly to clock input of Q0 flip-flop
3. $\overline{Q0}$ O/P given to CLK input of Q1 flip-flop, $\overline{Q1}$ o/p given to CLK input of Q2 flip-flop
4. Count Enable = 1, during Counting (Count Enable = 0, disables the counting)
5. Total States of the counter = 8, Hence Mod-8 Counter
6. As the count sequence is 7->6->5->4->3->2->1->0->7...it is binary down counter

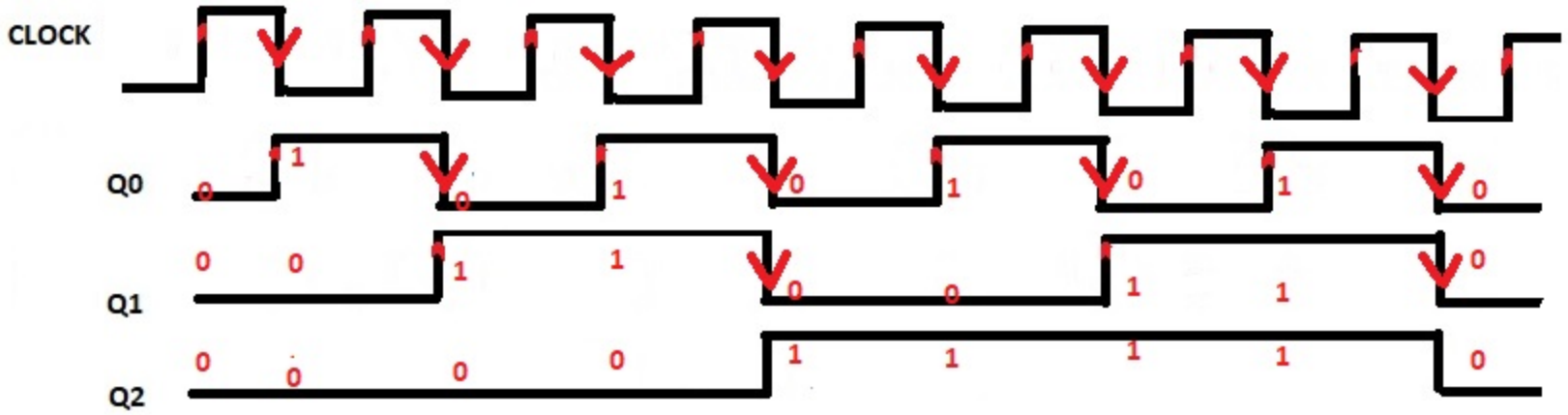
Mod-8 Asynchronous Counter for UP Counting Using -Ve Edge Triggered T Flip-Flops and Using Q o/p



COUNT SEQUENCE

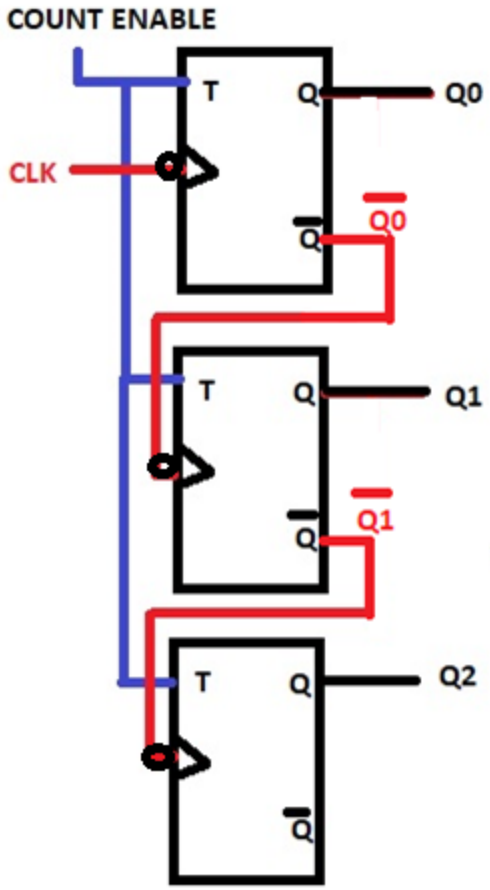
	Q2	Q1	Q0
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1	1	1
	0	0	0

MOD - 8 AYNCHRONOUS COUNTER



1. -VE Edge Triggered T flip-flops are used
2. Input clock is given directly to clock input of Q0 flip-flop
3. Q0 O/P given to CLK input of Q1 flip-flop, Q1 o/p given to CLK input of Q2 flip-flop
4. Count Enable = 1, during Counting (Count Enable = 0, disables the counting)
5. Total States of the counter = 8, Hence Mod-8 Counter
6. As the count sequence is 7->6->5->4->3->2->1->0->7...it is binary down counter

Mod-8 Asynchronous Counter for Down Counting Using -VE Edge Triggered T Flip-Flops ,Using Complimentary o/p



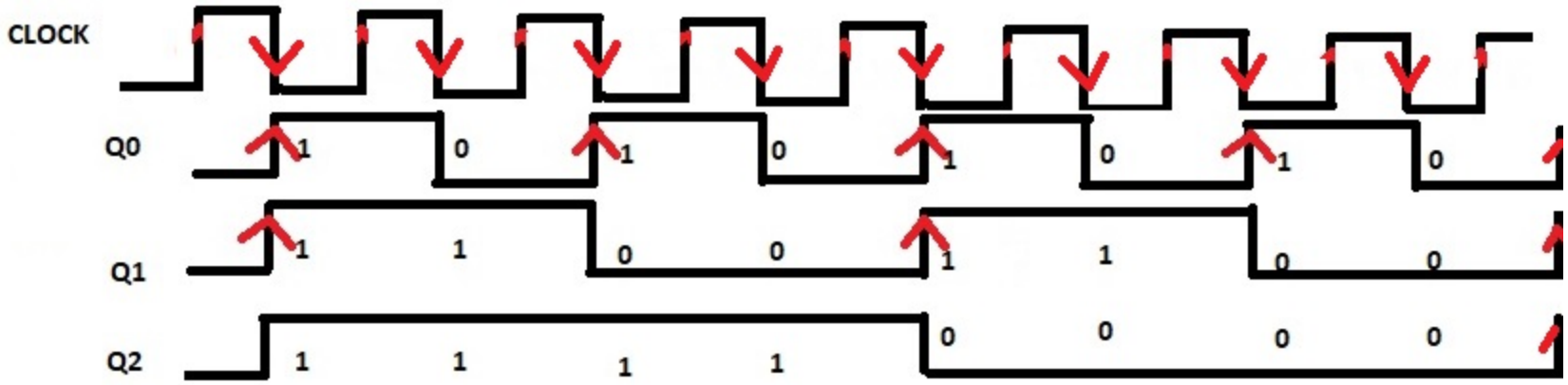
COUNT SEQUENCE

	Q2	Q1	Q0
7	1	1	1
6	1	1	0
5	1	0	1
4	1	0	0
3	0	1	1
2	0	1	0
1	0	0	1
0	0	0	0
	1	1	1

MOD - 8 AYNCHRONOUS COUNTER

Down Counter

Using -ve Edge Triggered T flip-flops (used $\overline{Q}$ outputs)



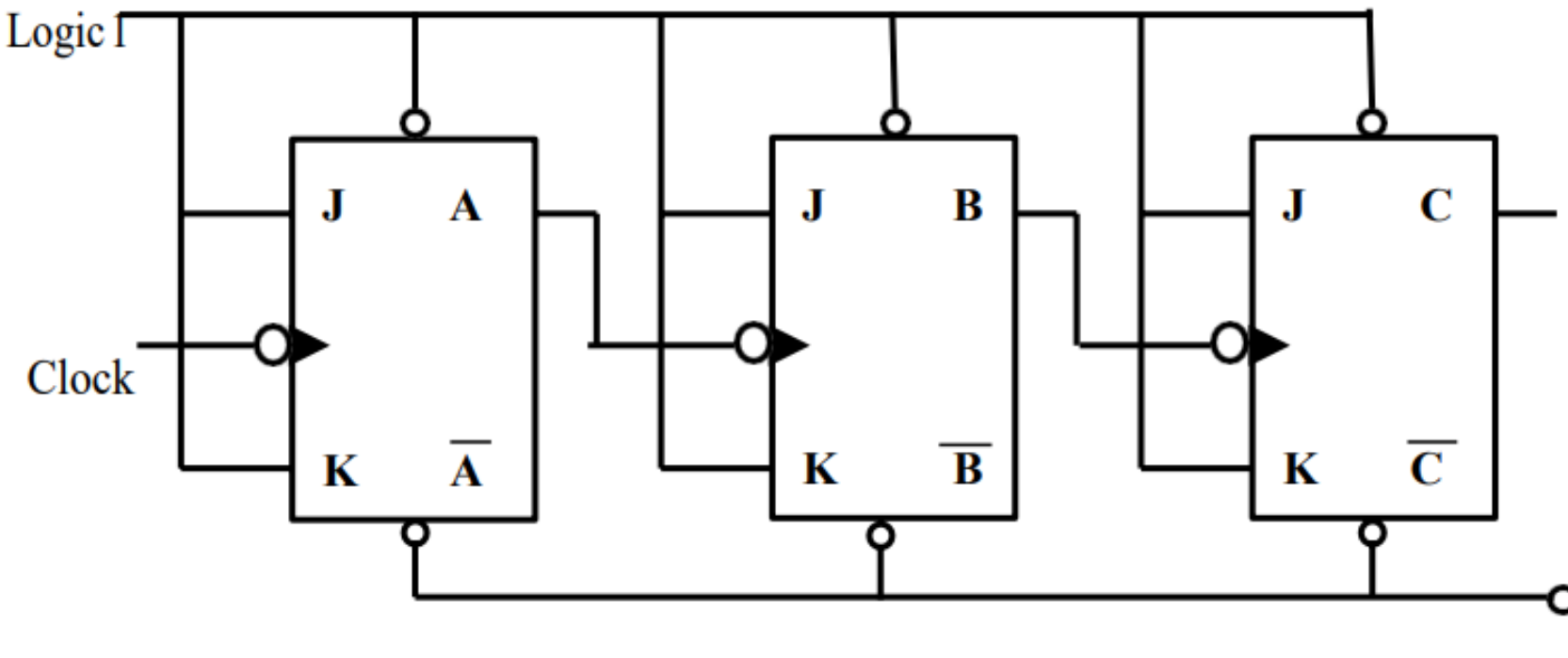
1. -VE Edge Triggered T flip-flops are used
2. Input clock is given directly to clock input of Q0 flip-flop
3. $\overline{Q0}$ O/P given to CLK input of Q1 flip-flop, $\overline{Q1}$ o/p given to CLK input of Q2 flip-flop
4. Count Enable = 1, during Counting (Count Enable = 0, disables the counting)
5. Total States of the counter = 8, Hence Mod-8 Counter
6. As the count sequence is 7->6->5->4->3->2->1->0->7...it is binary down counter

We can easily design a counter which counts through 2,4,8,16,32 by using required number of flip-flop. But often counters other than module, 2,4,8,16,32 etc are required.

For example : Mod7, mod6, mod5 etc. These counters can be designed by nearest highest modulus counter by skipping some of the counts.

Describe operation of Mod 7 or divide by 7 counter Ans.

Mod7 counter can be designed from mod8 counter by skipping one of the state. Similarly a mod6 counter can be designed from mod 8 counter by skipping two of the states.

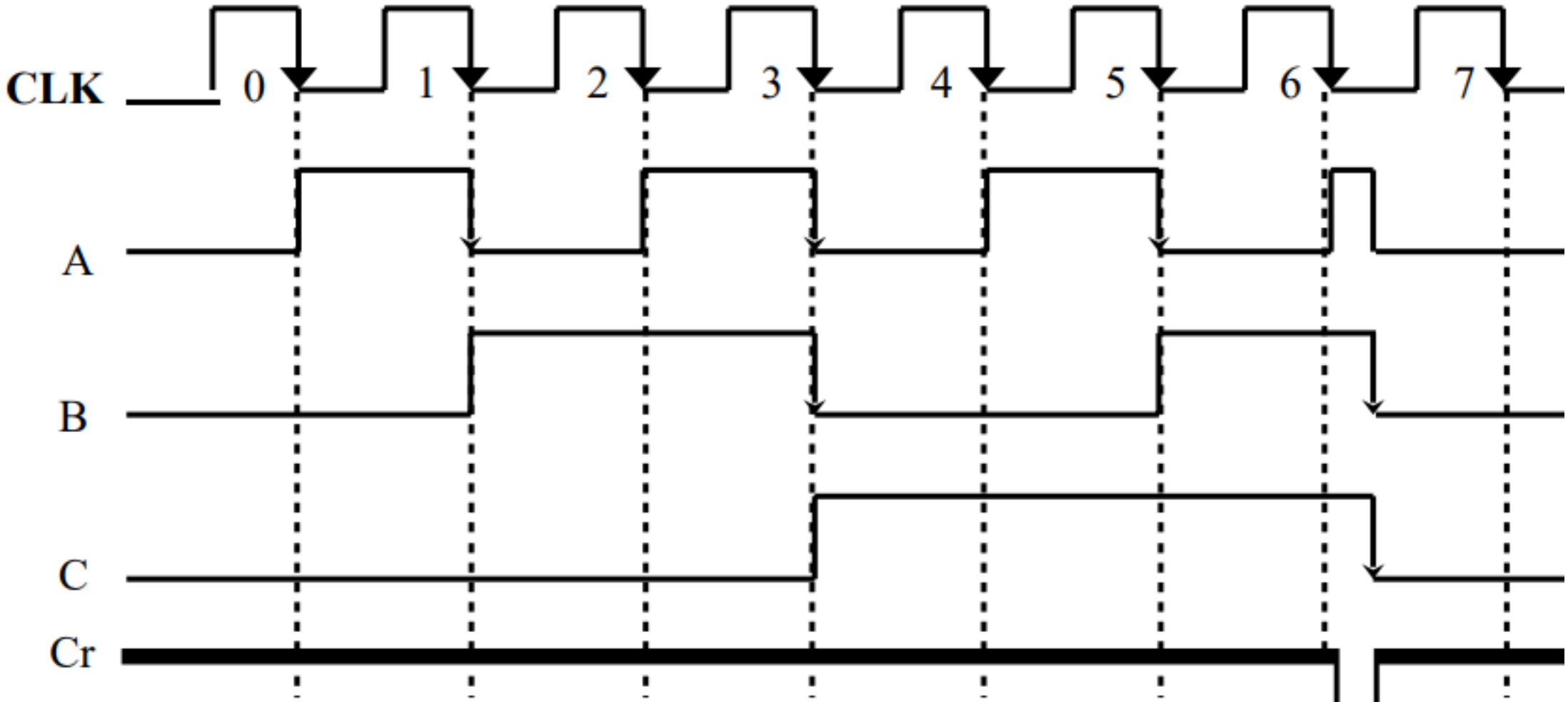


Truth table

Clock	C	B	A
0	0	0	0
1	0	0	1
2	0	1	0
3	0	1	1
4	1	0	0
5	1	0	1
6	1	1	0
7	1(0)	(0)	1(0)

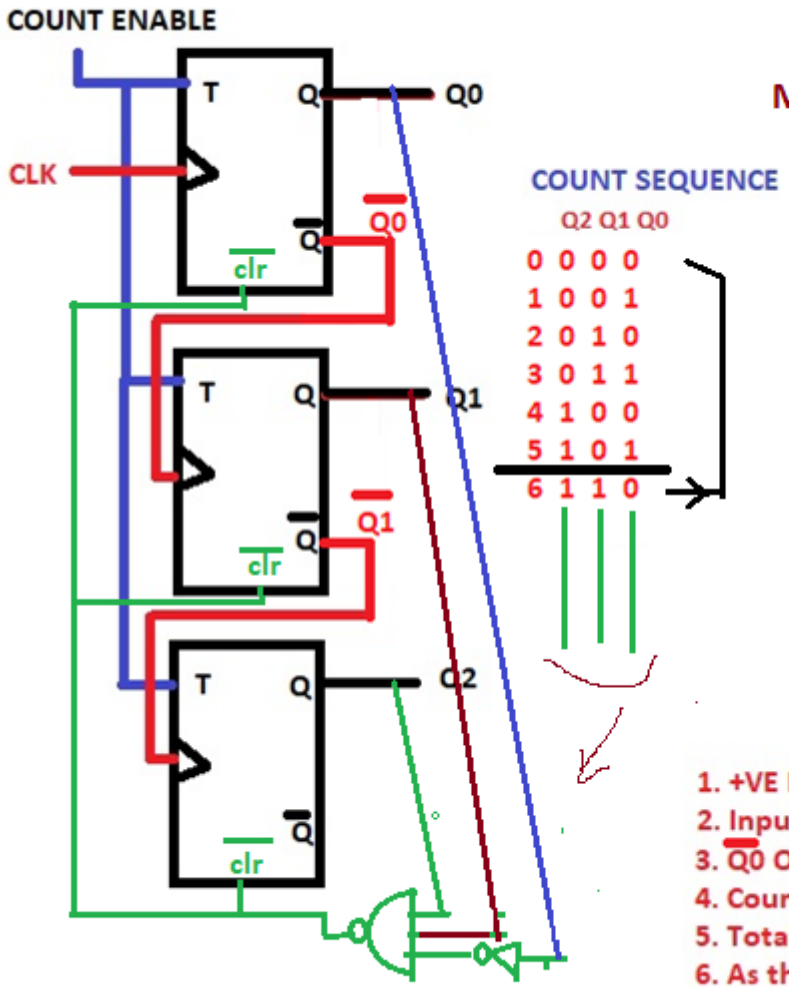
Fig : Mod-7 Counter

WaveForms :



Count	000	001	010	011	100	101	110	000
State	0	1	2	3	4	5	6	7(0)

Mod-6 Asynchronous Counter Using +VE Edge Triggered T Flip-Flops



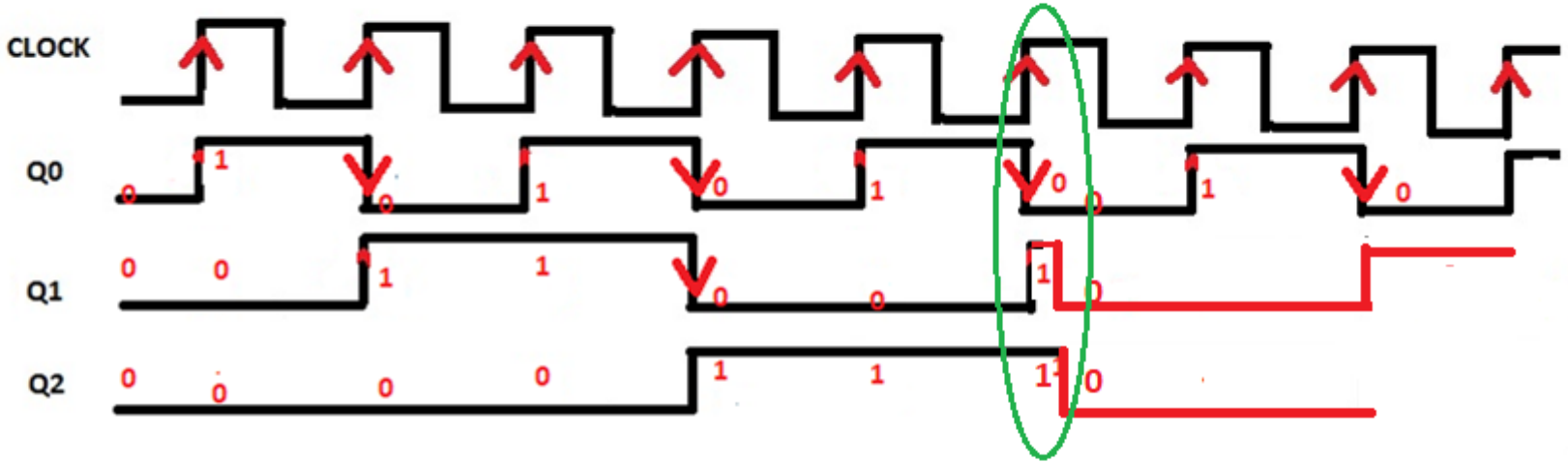
COUNT SEQUENCE

Q2 Q1 Q0
0 0 0 0
1 0 0 1
2 0 1 0
3 0 1 1
4 1 0 0
5 1 0 1
6 1 1 0

MOD6 AYNCHRONOUS COUNTER

Mod 6 Up CounterUsing +ve Edge Triggered flip-flops

NAND gate identifies the 110 condition and drives all the o/ps to 000, by using asyncronous clr input..again seqence begins



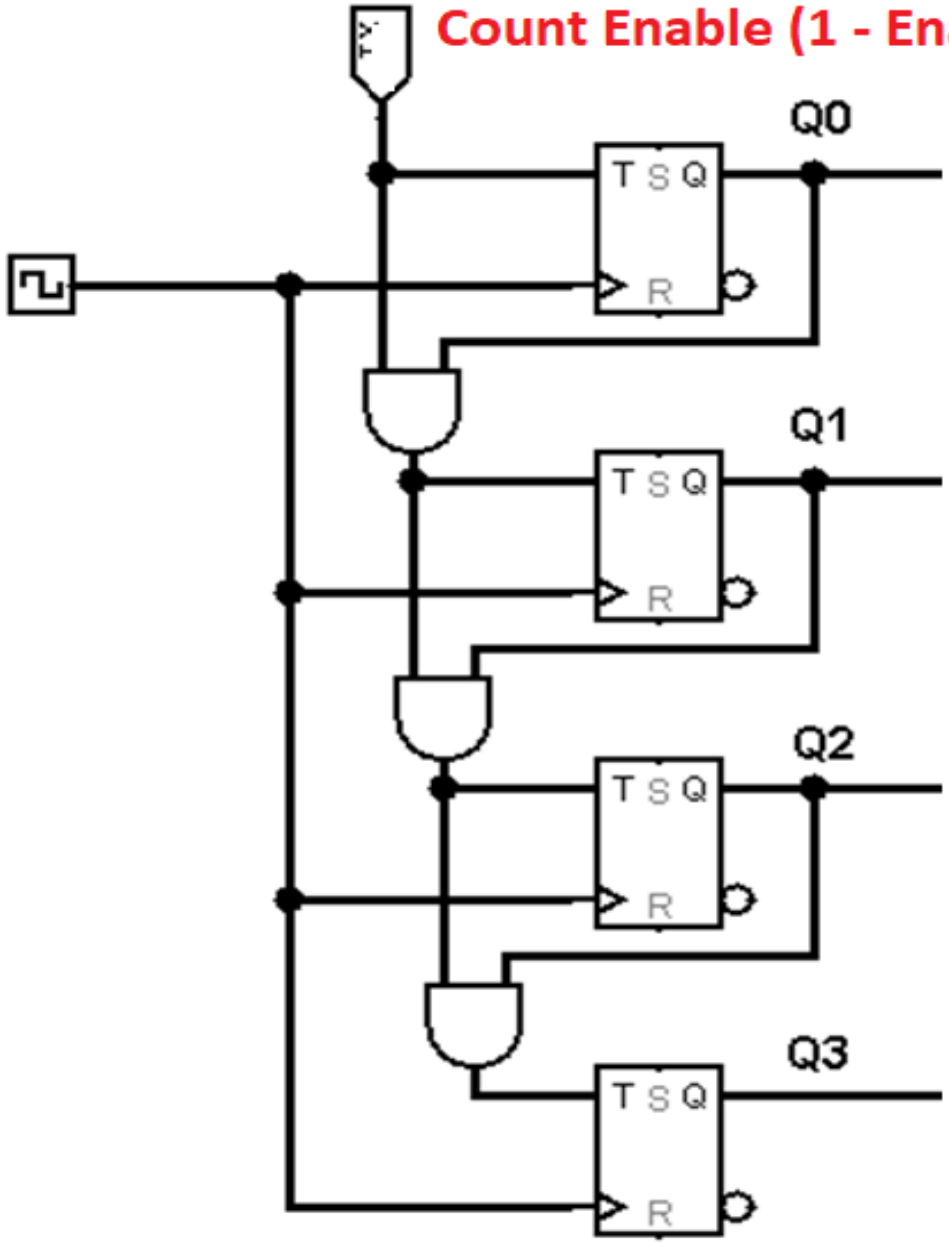
1. +VE Edge Triggered T flip-flops are used
2. Input clock is given directly to clock input of Q0 flip-flop
3. $\overline{Q0}$ O/P given to CLK input of Q1 flip-flop, $\overline{Q1}$ o/p given to CLK input of Q2 flip-flop
4. Count Enable = 1, during Counting (Count Enable = 0, disables the counting)
5. Total States of the counter = 8, Hence Mod-8 Counter
6. As the count sequence is 7->6->5->4->3->2->1->0->7...it is binary down counter

Ripple counters suffer from slow speeds, cumulative propagation delays (rippling effect), and potential counting errors (glitches) at higher frequencies because each flip-flops clock input depends on the previous ones output, making them less reliable than synchronous counter for complex tasks or high speed operations.

Key Drawbacks:

1. Propagation delay & Speed Limitation : The cumulative delay restricts the maximum operating frequency, f(max), as the clock period must be longer than the total delay
   T(clock) > n x tpd,
   where tpd is the propagation delay of one flip-flop, n is no. of flip-flops
   Hence, highest frequency = 1 / n x tpd
2. Asynchronous Nature: The asynchronous clocking (each FF triggered by the previous output) makes them less predictable and harder to synchronize with other parts of a digital system compared to synchronous counters.
3. Natural counting sequence is pure binary, circuit becomes complex and speed decreases for other counting sequences like BCD etc.

4 BIT SYNCHRONOUS BINARY COUNTER (MOD 16 COUNTER / %16 COUNTER)



Q3	Q2	Q1	Q0
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	1	1	1

Q1 TOGGLE FOR EVERY CLK

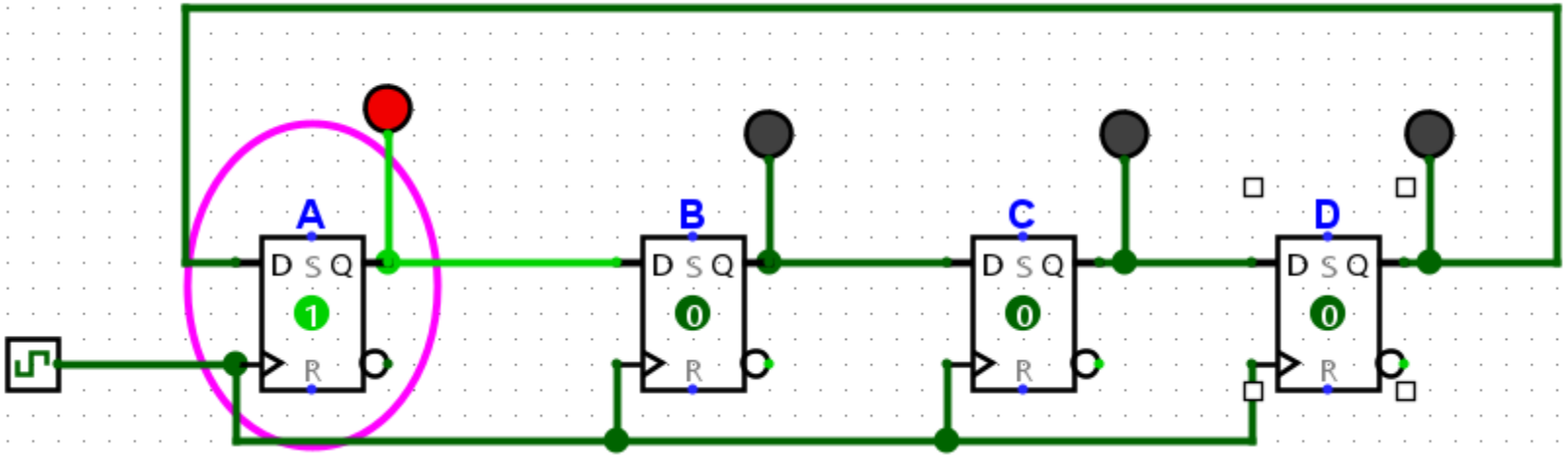
Q1 TOGGLE WHEN Q0 IS 1

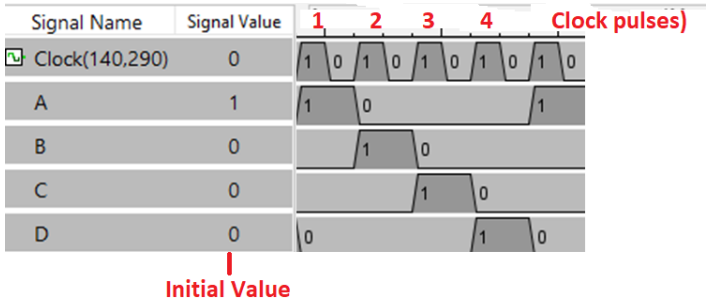
Q2 TOGGLE WHEN Q0,Q1 IS 1

Q3 TOGGLE WHEN Q0,Q1,Q2 IS 1

A **ring counter** (also called as One-Hot Counter) is a type of digital counter made from a circular shift register where the output of the last flip-flop is fed back into the input of the first flip-flop. **Structure:** It uses n D flip-flops for an n-state counter, sharing a common clock signal. A single 1 (high) bit is preset into one flip-flop while the rest are set to 0 (low). **Rotation:** With each clock pulse, the 1 bit shifts sequentially from one flip-flop to the next in a continuous loop. For an n-bit ring counter, it cycles through n unique states.

Common Applications : **State Sequencers:** Controlling step-by-step or timed operations in finite-state machines, **Frequency Division:** Dividing a clock frequency by the number of stages in the ring, **Timing & Display Signals:** Generating multi-phase square waves, quadrature signals, or driving step-sequenced displays and stepper motors.





Clock pulse	Q_A	Q_B	Q_C	Q_D
0	1	0	0	0
1	0	1	0	0
2	0	0	1	0
3	0	0	0	1
4	1	0	0	0

Table 6.6.1 Ring counter sequence 4-bits

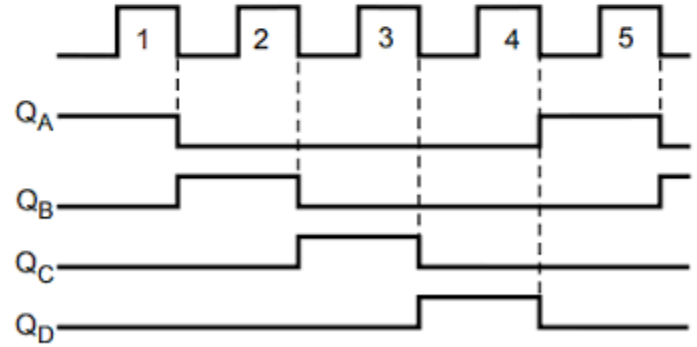
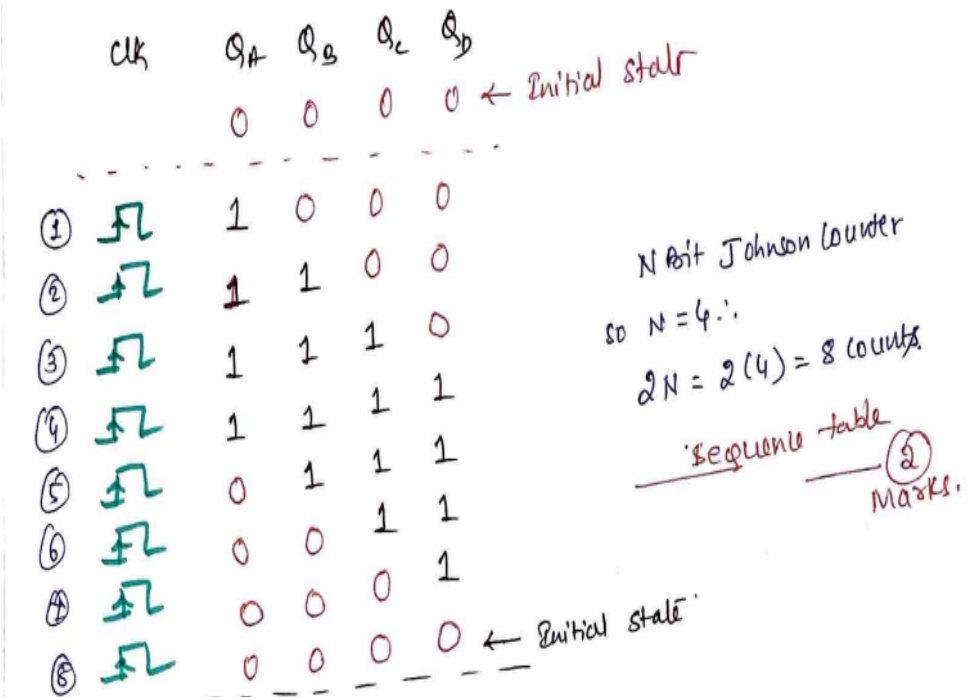
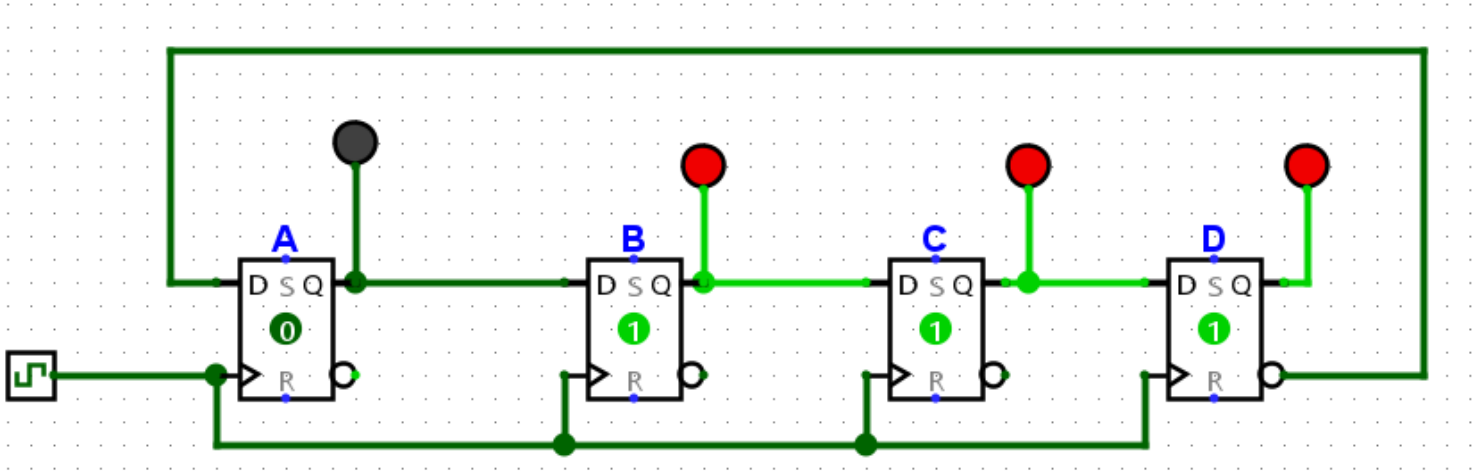
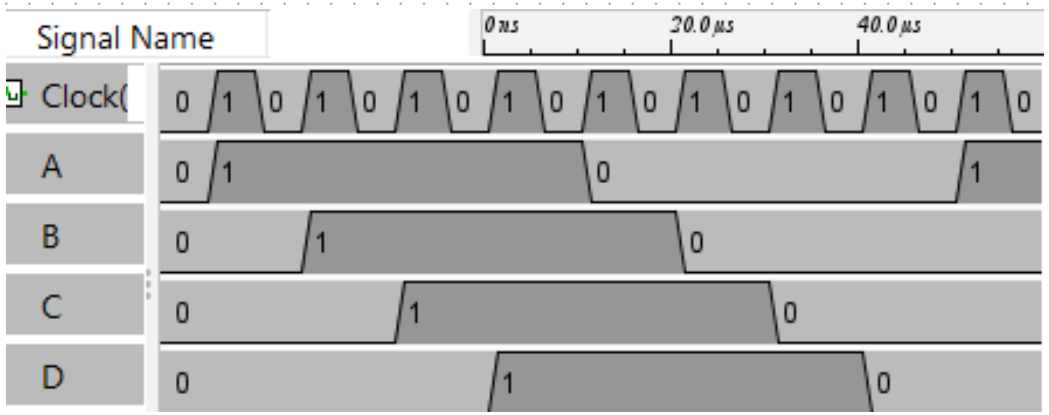


Fig. 6.6.2 Timing sequence for a four-bit ring counter

Johnson Counter (Twisted Ring Counter): The *complemented* (inverted) output of the last flip-flop loops back to the first flip-flop. It circulates a stream of ones followed by zeros and provides 2n states. A **Johnson counter** also called a twisted ring counter, switch-tail counter, or Möbius counter is a synchronous shift register circuit where the inverted output of the last flip-flop is fed back into the input of the first flip-flop. **Number of States:** An n-bit Johnson counter generates 2n unique states (a MOD-2n counter). For example, a 4-bit counter uses 4 flip-flops to produce 8 states. Counts twice the number of states (2n) as a standard ring counter (n) using the same number of flip-flops. **Gray Code Output:** Adjacent states differ by only a single bit (Hamming distance of 1), preventing glitches during asynchronous sampling. **Self-Decoding:** Individual states can be easily decoded using simple logic gates. Common Applications: **Frequency Dividers:** Dividing a clock frequency by 2n, **Phase Generators:** Creating multi-phase square wave generators, **Finite State Machines (FSM):** Used in ASIC and FPGA design for control and sequencing







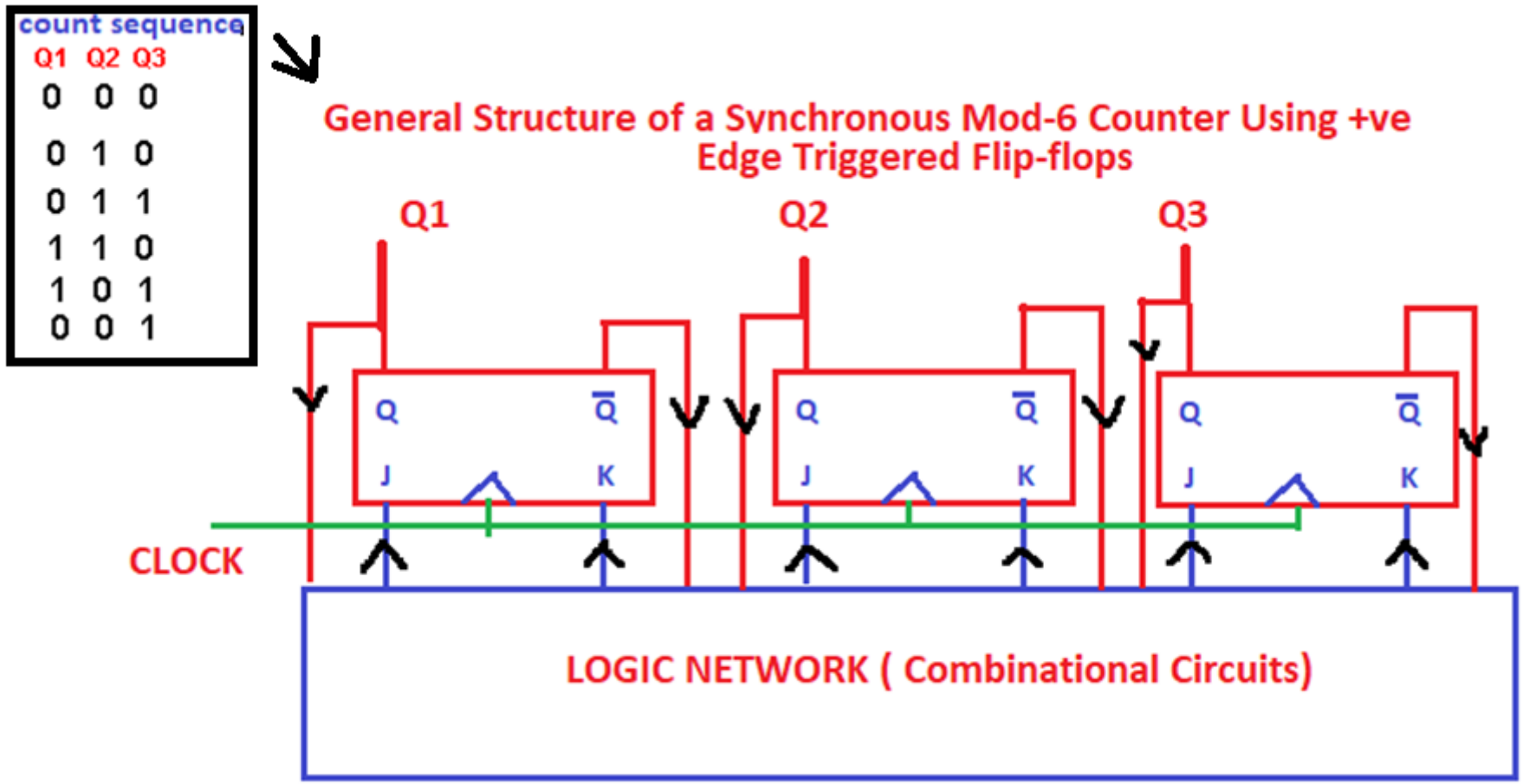
Ripple counters suffer from slow speeds, cumulative propagation delays (rippling effect), and potential counting errors (glitches) at higher frequencies because each flip-flops clock input depends on the previous ones output, making them less reliable than synchronous counter for complex tasks or high speed operations.

Key Drawbacks:

1. Propagation delay & Speed Limitation : The cumulative delay restricts the maximum operating frequency, f(max), as the clock period must be longer than the total delay
   T(clock) > n x tpd,
   where tpd is the propagation delay of one flip-flop, n is no. of flip-flops
   Hence, highest frequency = 1 / n x tpd
2. Asynchronous Nature: The asynchronous clocking (each FF triggered by the previous output) makes them less predictable and harder to synchronize with other parts of a digital system compared to synchronous counters.
3. Natural counting sequence is pure binary, circuit becomes complex and speed decreases for other counting sequences like BCD etc.

Design a Mod-6 counter for the given counting sequence using JK Flip-flops:

3 JK flip-flops are used to realize the counter. Design principle is shown in the figure, The current state of the counter is applied to a logic network. The function of the logic network is to generate the appropriate Signals for the J and K terminals of the clocked flip-flops so that the specified next state in the counting sequence results upon the occurrence Of a triggering edge of the clock input. The design is carried out by preparing Excitation Table (Present state,Next state, JK ff inputs) and generating minimal sum expressions for all the JK inputs.



Sychronous Mod-6 Counter (custom sequence) using JK Edge Triggered Flip-flops

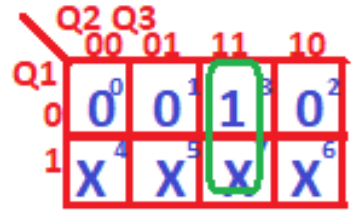
Excitation Table

Present State Q1 Q2 Q3	Next State Q1+ Q2+ Q3+	Flip-Flop Inputs (derived from Application Table) J1 K1	J2 K2	J3 K3
0 0 0	0 1 0	0 X	1 X	0 X
0 1 0	0 1 1	0 X	X 0	1 X
0 1 1	1 1 0	1 X	X 0	X 1
1 1 0	1 0 1	X 0	X 1	1 X
1 0 1	0 0 1	X 1	0 X	X 0
0 0 1	0 0 0	0 X	0 X	X 1

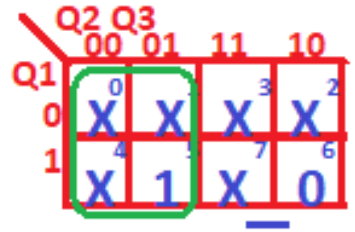
Q	Q+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

Application Table for JK Flip-Flops

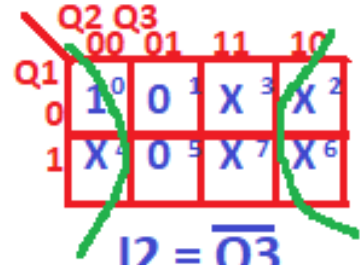
Determination of Minimal sum expressions for JK flip flop inputs as per excitation table



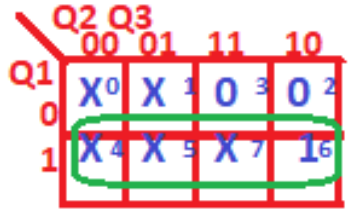
$J1 = Q2.Q3$



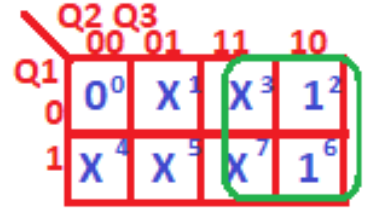
$K1 = \overline{Q2}$



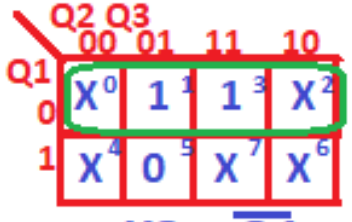
$J2 = \overline{Q3}$



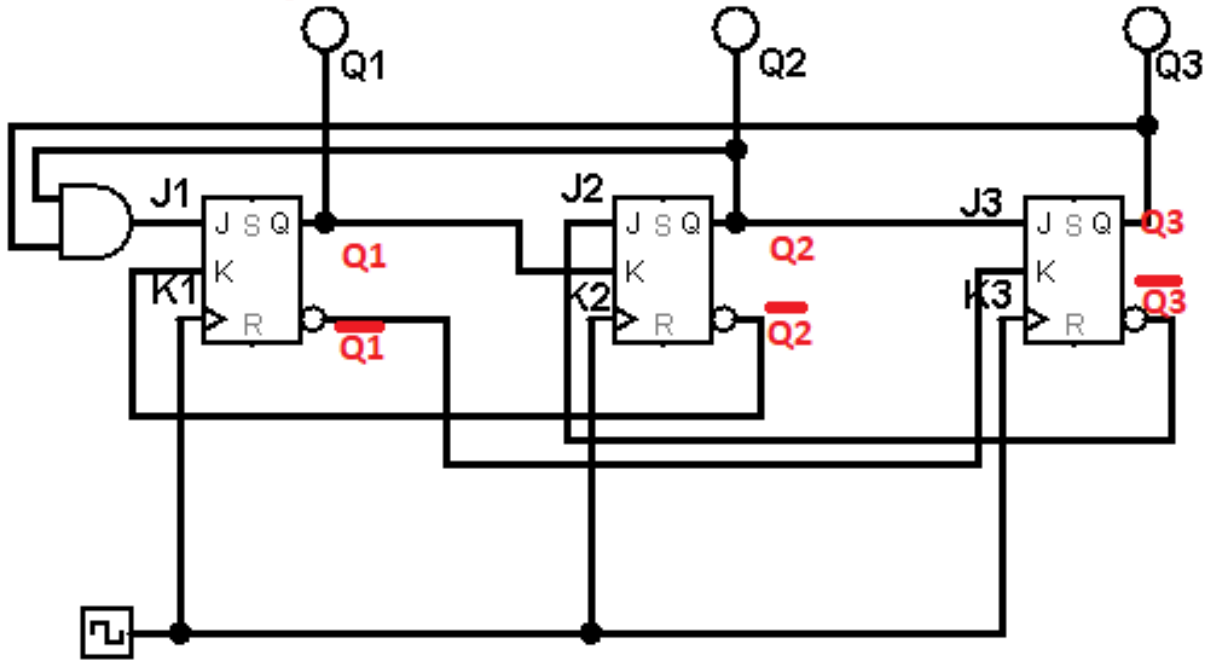
$K2 = Q1$



$J3 = Q2$



$K3 = \overline{Q1}$



Logic Diagram of Synchronous Mod-6 Counter (as per the derived minimal expressions)

count sequence

Q1	Q2	Q3
0	0	0
0	1	0
0	1	1
1	1	0
1	0	1
0	0	1

Self Correcting Counters…

The counting sequence of mod-6 counter did not include states 100 and 111. Once the counter is Designed, definite next states result if either of these states should occur.

The reasons for the counter to enter invalid states:
When power is first applied to the network, the initial states of flip-flops may be unpredictable
Noise signals, can cause the counter to enter in to any one of the invalid states

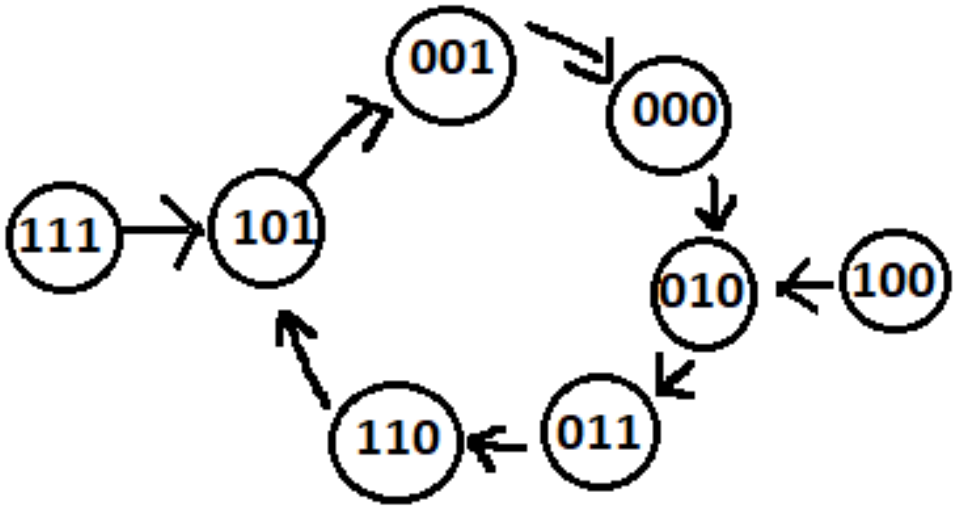
A counter in which all the states not included in the original counting sequence eventually lead to the Normal counting sequence after one or more clock pulses are applied, is said to be Self Correcting.
To avoid having the counter "hang up", (lock out), it should always be designed as self-correcting.

Let us test the previous design is self correcting ?
Not used States: 100 AND 111. Let us verify, when these states occur what will be the next state

Q1	Q2	Q3	J1 K1 (q2.q3,*q2)	J2 K2 (*q3,q1)	J3 K3 (q2,*q1)	Q1+	Q2+	Q3+	Remarks
1	0	0	0,1	1,1	0,0	0	1	0	leads to VALID state
1	1	1	1,0	0,1	1,0	1	0	1	leads to VALID state

Hence, counter designed is a SELF CORRECTING COUNTER

State Diagram



Counters having no unused states are always self correcting. It is the assignment to the don’t cares associated with the unused States, when the logic expressions are obtained that can cause a counter not to be self correcting.

By actually specifying the next states for each of the unused states in the counting sequence prior to constructing the Kmaps, a self correcting counter realization can be guaranteed.

5.a Sequence generator.

$0 \to 1 \to 2 \to 4 \to 5 \to 6 \to 0 \to 1 \to 2$

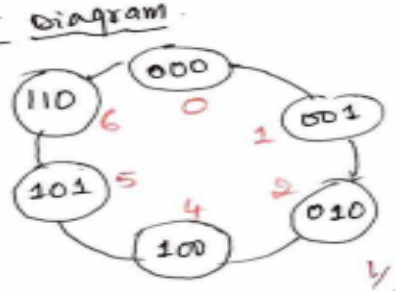
6 → is the Highest count ∴ we need (3) Flip Flop to construct the above sequence.

JK Excitation table

Q	Q^+	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

— 1/2 Mark

State diagram

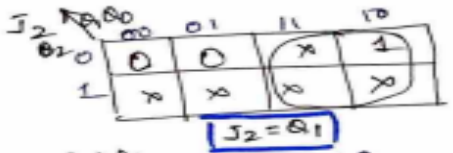


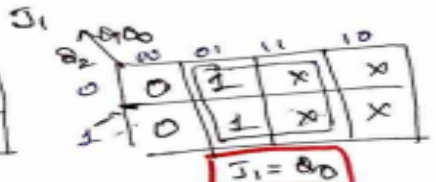
1/2 Mark

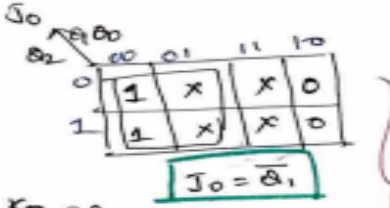
State table

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	J_2 K_2	J_1 K_1	J_0 K_0
0	0	0	0	0	1	0 x	0 x	1 x
0	0	1	0	1	0	0 x	1 x	x 1
0	1	0	1	0	0	1 x	x 1	0 x
0	1	1	x	x	x	x x	x x	x x
1	0	0	1	0	1	x 0	0 x	1 x
1	0	1	1	1	0	x 0	1 x	x 1
1	1	0	0	0	0	x 1	x 1	0 x
1	1	1	x	x	x	x x	x x	x x

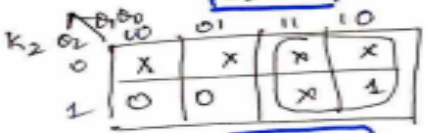
1 1/2 Marks

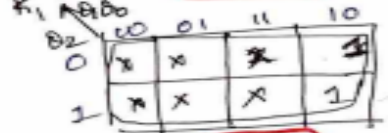


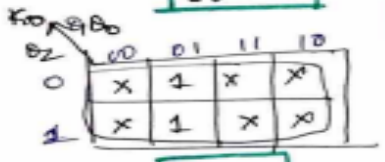


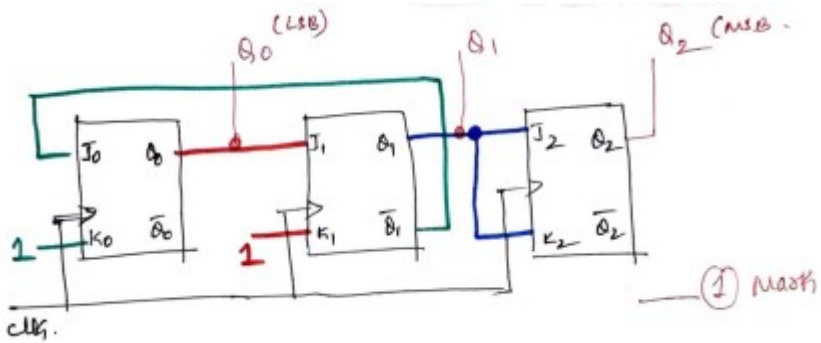


1 1/2 Marks









(1) Mark

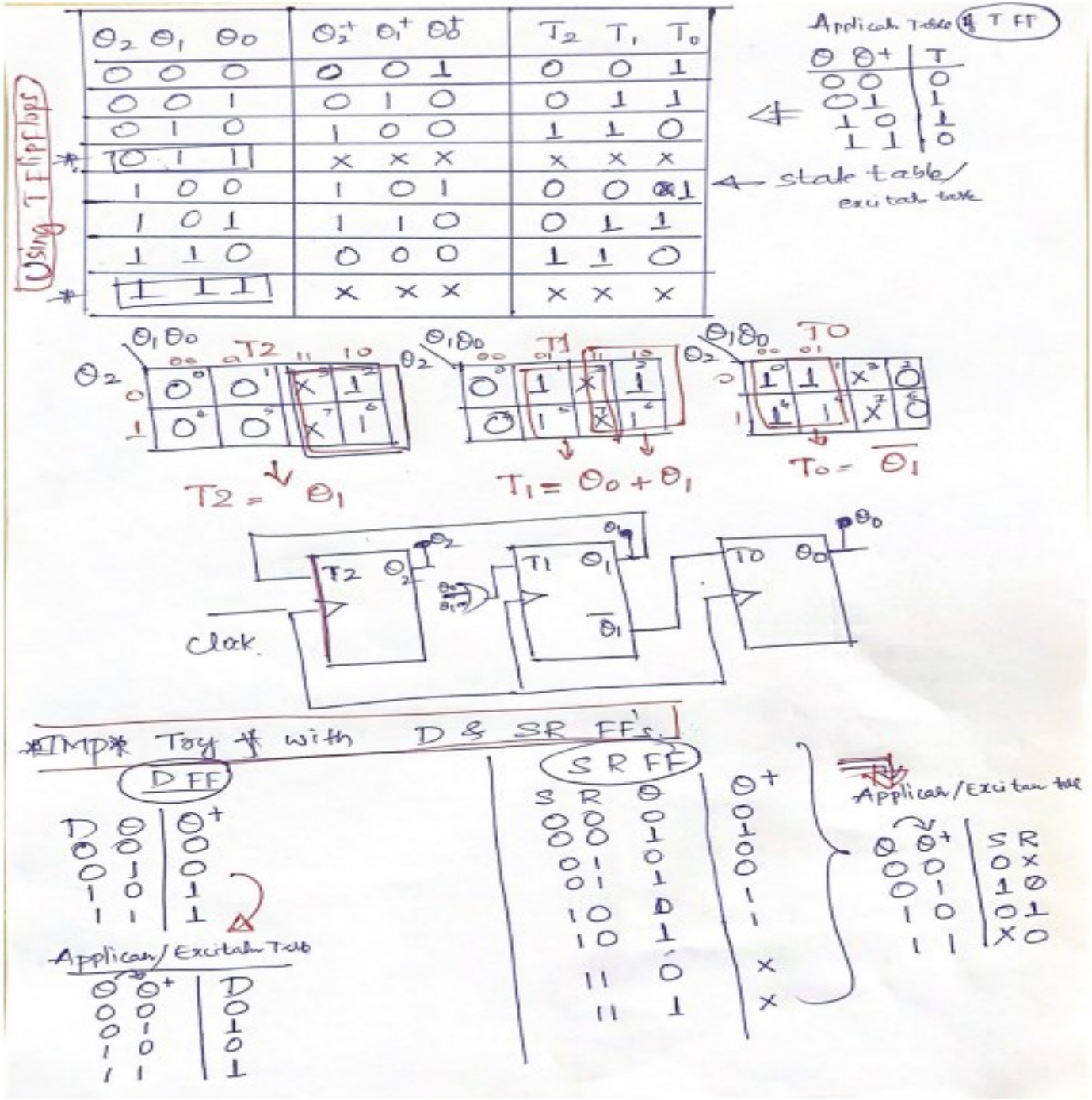
Only if asked you can write, otherwise not required

Nok

$Q_2Q_1Q_0$	J_2K_2	J_1K_1	J_0K_0	Q_2^+	Q_1^+	Q_0^+	
0 1 1	$(Q_1,\bar{Q}_1)$	$(Q_0,1)$	$(\bar{Q}_1,1)$				
	(1,1)	(1,1)	(0,1)	1	0	0	(4)
1 1 1	(1,1),	(1,1)	(0,1)	0	0	0	(0)

When Invalid states 3 & 7 occured, the next state will be a Valid state (4 & 0). Hence, it is a self correcting Counter

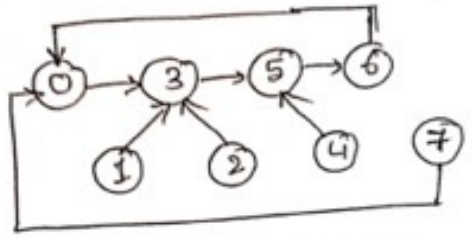
Using T Flip Flops



Design of a synchronous counter for the following count sequence: 0, 3, 5, 6 using T flip flop, (Avoid Lock out condition by forcing unused states to next counting state). CIE, 2025

6) Synchronous Counter: Using T-FF

Need 3-T-FF



⇒ unused state

State Table — 4 Marks

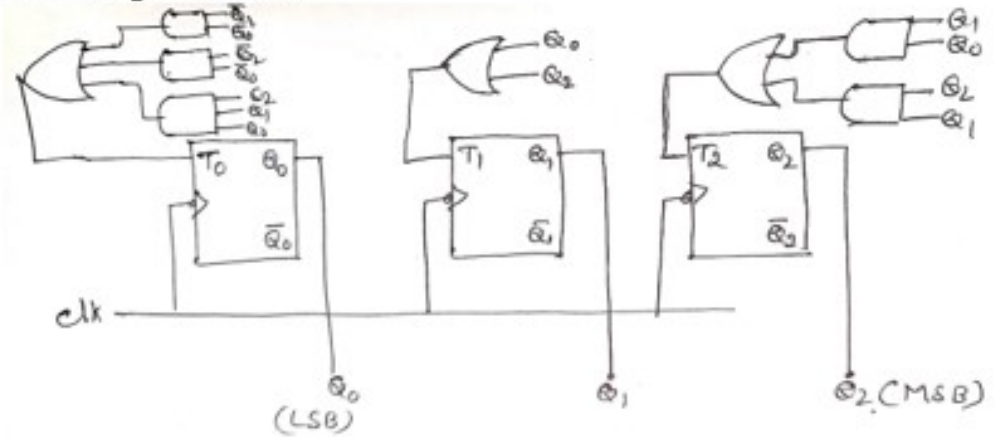
P. S.			Next state			Inputs		
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	T_2	T_1	T_0
0	0	0	0	1	1	0	1	1
0	0	1	0	1	1	0	1	0
0	1	0	0	1	1	0	0	1
0	1	1	1	0	1	1	1	0
1	0	0	1	0	1	0	0	1
1	0	1	1	1	0	0	1	0
1	1	0	0	0	0	1	1	0
1	1	1	0	0	0	1	1	1

FF inputs: Use K-map — 3 Marks

$T_2 = Q_1Q_0 + Q_2Q_1$

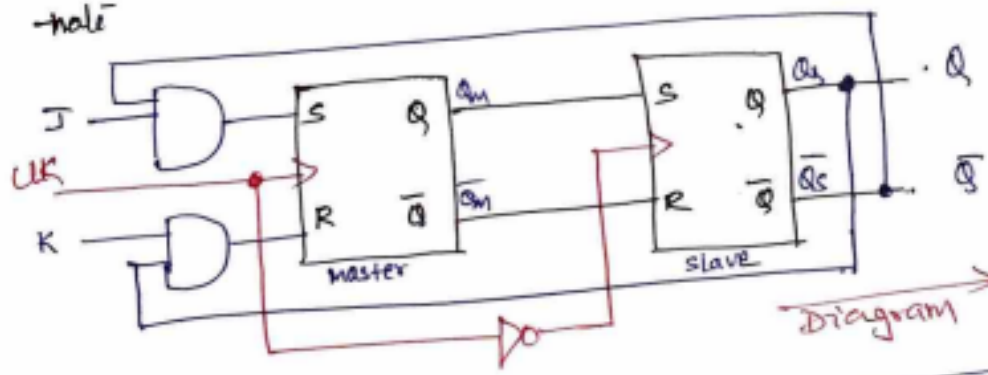
$T_1 = Q_0 + Q_2$

Circuit Diagram:--3 Marks

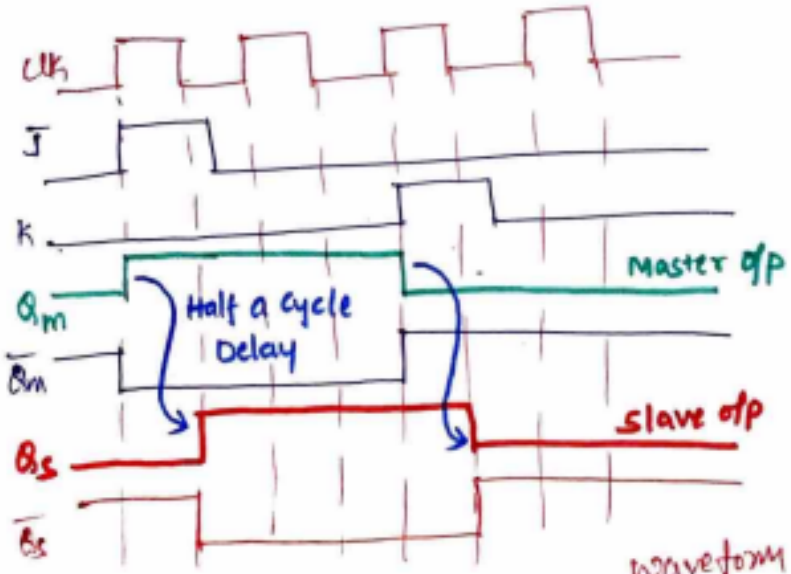


Explain the phenomenon of a race-around condition in sequential circuits, particularly in JK flip-flops, and describe how this issue is mitigated using a master-slave JK flip-flop configuration. Additionally, illustrate the use of a gated SR latch to control the input signals, providing a detailed logic diagram and timing waveforms to demonstrate the elimination of the race-around condition in the master-slave configuration. (CIE,2024)

A.a Race around means continuous toggling. The clock pulse is HIGH for a longer duration than the Flipflop propagation delay, this causes the output to toggle continuously within the same clock, making it Indeterminate state



(2) Marks Diagram



waveform (2) Marks

CLK	J	K	Q	$\bar{Q}$	State
⎍	0	0	Q	$\bar{Q}$	NO cha
⎍	0	1	0	1	Reset
⎍	1	0	1	0	Set
⎍	1	1	$\bar{Q}$	Q	toggle
0	x	x	Q	$\bar{Q}$	NO chang

Truth table (1) Mark

SEE, 2024

1) Design a Mod 4 synchronous up-counter using JK flipflop. (ref prev slides)

2) Design a synchronous 7,4,3, 1,5, 0,7, 4, 3, counter to count the following sequence ... implement the circuit using T-flipflops. (ref prev slides)

3) Draw the timing diagram of a 2 bit down counter with negative edge triggered flip flops. (refer prev slides)

4) How many JK flipflops are required to achieve the frequency division of 8? (each FF is %2, hence 4)

CIE2,2025

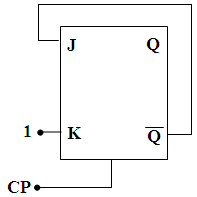
How many flip-flops are required to design a Mod-20 counter?

Ans: 5 FF

Given the present state = 0 and next state = 1 for a JK FF, find the required values of J and K inputs.

Ans: J=1; K=X

A JK flip-flop is configured as shown in the figure. The flip-flop is initially reset (Q = 0) and is then clocked for six consecutive pulses. Determine the output sequence at Q for the six clock pulses.



Ans: 101010

A flip-flops has a 3 ns delay. What is the maximum delay in a 10-bit binary ripple counter that uses these flip-flops? What is the maximum frequency at which the counter can operate?

Ans: Total Delay=10x3 ns= 30 ns

Fmax=1/30ns=33.33MHz

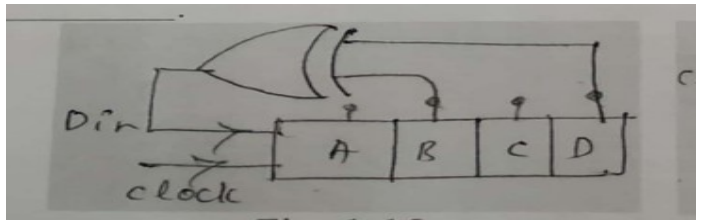
CIE,2025

A binary counter has 4 outputs, Q3, Q2, Q1, Q0, with Q3 as MSB. A NAND gate output is connected to active low clear inputs of all Flip-flops. What is the last count one can observe if Q3 and Q2 are connected as inputs to the NAND gate?

Ans: Last Count= 1011(Binary)
(11 in Decimal)

A 4 bit shift register circuit configured for right shift operation is shown below, if the present state of the shift register is ABCD = 1101, then the number of clock cycles required to reach the state ABCD = 1111 is ----------

Ans: 9 clock pulses



Describe the working principle of a JK master–slave flip-flop with suitable waveforms. Explain why the JK master–slave flip-flop eliminates the race-around condition.

Ans:

The Master-Slave configuration eliminates race-around condition by splitting the flip-flop into two stages that respond to opposite phases of the clock signal. Clock Pulse High: The Master is enabled. It accepts the J and K inputs and "decides" what the next state should be. However, the Slave remains disabled (locked), so the final output Q does not change yet. Since Q hasn't changed, the feedback sent back to the Master's inputs remains constant. This prevents the "looping" toggle that causes the race condition. (ref slides/ref book)

Derive characteristic equation for JK and D Flip-flop and show how the D flip flop can be converted to a JK flip flop.

Discuss the concept of propagation delay in ripple counters and interpret how this delay affects the maximum operating frequency. Design Negative (↓) Edge triggered MOD 6 Asynchronous up counter using T flip flop.

Ans:

Propagation Delay is the time delay between the moment a signal transition occurs at the input (like the clock edge) and the moment the resulting transition appears at the output. Every flip-flop has an inherent, non-zero propagation delay dictated by the physical switching speed of its internal transistors. In a ripple counter, the state change has to propagate through the entire chain of flip-flops sequentially. The accumulated delay directly dictates the speed limit of the ripple counter. For the counter to operate reliably, the next clock pulse must not arrive until all flip-flops have had sufficient time to settle to their new, correct state. Limiting Condition is the minimum time period allowed for the clock signal must be greater than or equal to the total accumulated propagation delay: --- 2 Marks

Instead of this you can use NAND gates, as explained in the prev slides

